

FLUIDS



General Science

Grade 11





What is a Fluid?



A **fluid** is any substance that can **flow** and continuously **change its shape** when subjected to an external force.

Unlike solids, fluids do not maintain a fixed shape. Instead, they **adapt to the shape** of the container or space that holds them.



Water takes the shape of its container.



Air expands to fill the space of the container.

There are two major types of fluids:



1. Liquids

Liquids have a definite volume but no fixed shape. They flow and take the shape of their container.



Examples:



Water



Milk



Cooking oil



Alcohol



2. Gases

Gases have neither a definite shape nor a definite volume. They expand to fill all available space.



Examples:



Air



Helium



Water vapor



Propane



Although liquids and gases have different properties, they are **both classified as fluids** because they can move freely and flow.



Why Are Liquids and Gases Both Called **Fluids**?

Liquids and gases are grouped together because they behave similarly in several ways.

Both:



can flow from one place to another



do not have a fixed shape



exert pressure



can move through pipes and channels

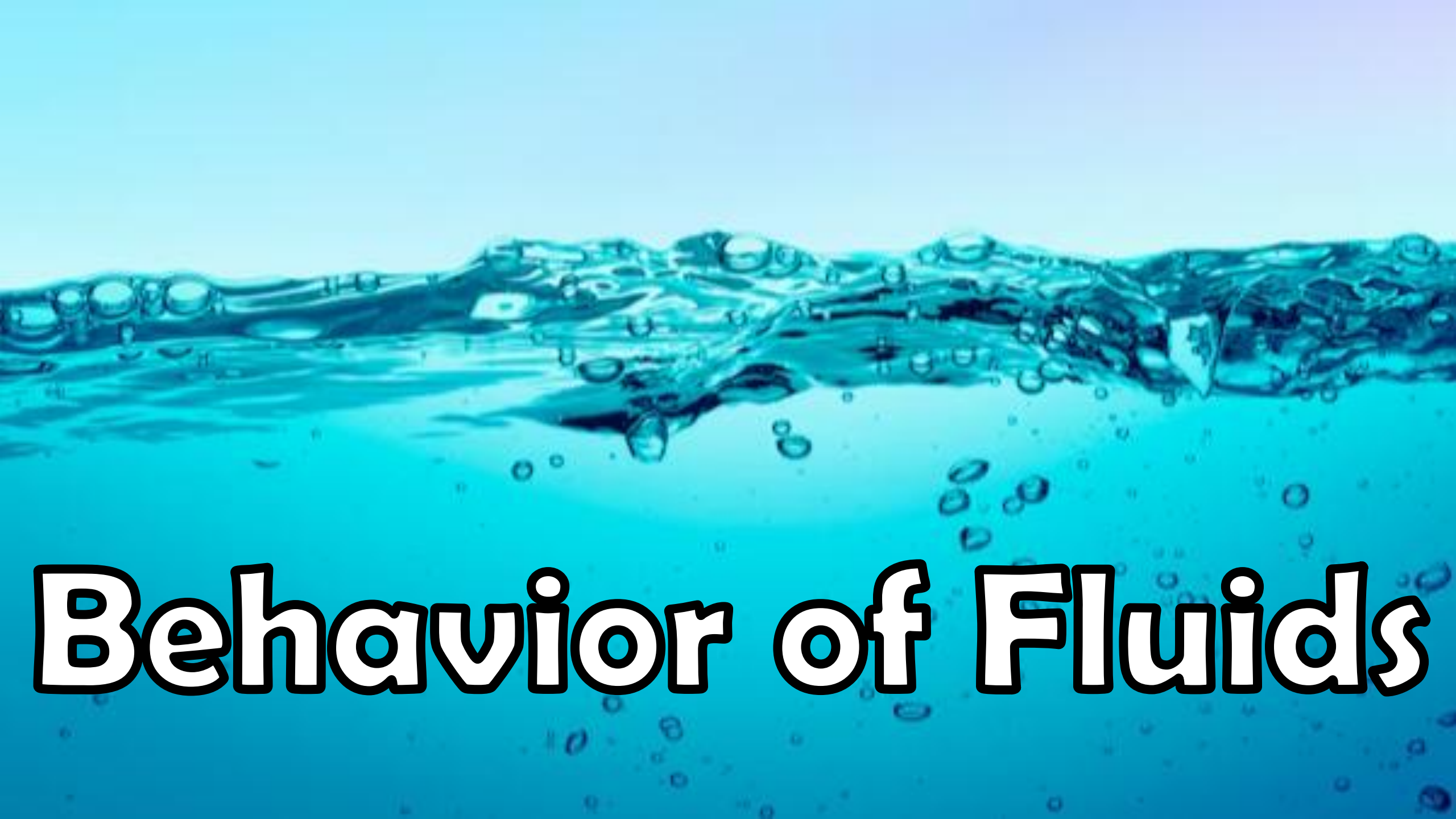


respond to applied forces



Although gases spread out more than liquids because their particles are farther apart, both are considered **fluids** because of their ability to flow.





Behavior of Fluids

1. Fluids Can Flow

The defining characteristic of every fluid is its ability to flow.

Flow occurs because the particles of a fluid are free to move past one another. This movement allows fluids to change shape continuously when acted upon by gravity or another force.

Examples:



Water flows through rivers.



Milk pours from a carton.



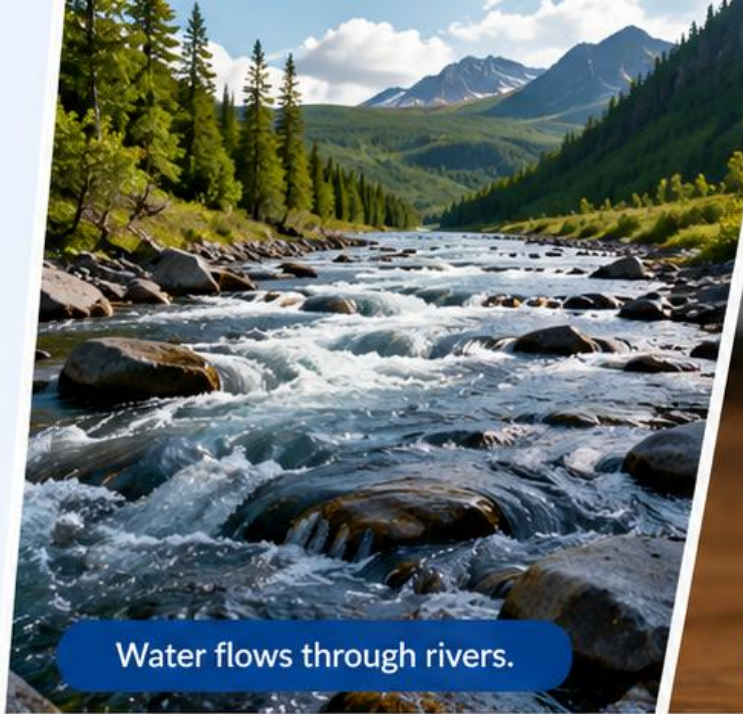
Air moves as wind.



Fuel travels through engine pipes.



Without this ability to flow, hydraulic systems, water distribution, and blood circulation would not be possible.



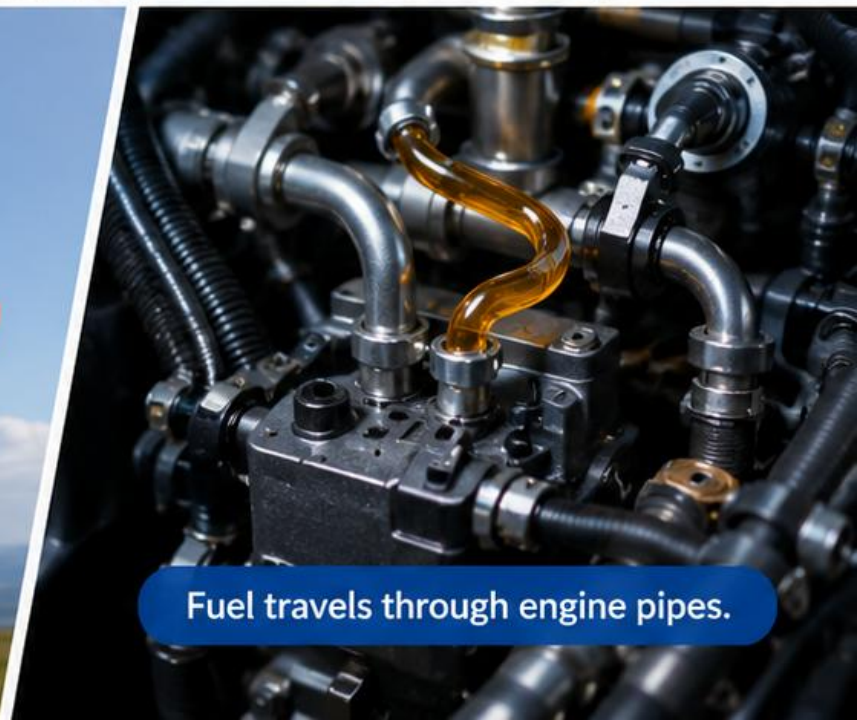
Water flows through rivers.



Milk pours from a carton.



Air moves as wind.



Fuel travels through engine pipes.

2. Fluids Take the Shape of Their Container



Unlike solids, fluids do not maintain their own shape.



Instead, they adjust to the shape of whatever contains them.

Examples:



Water appears tall inside a bottle but wide inside a bowl.



Cooking oil takes the shape of the bottle in which it is stored.



Air expands to fill an inflated balloon.



The shape changes, but the fluid itself remains the same substance.

Water appears tall inside a bottle but wide inside a bowl.



Same water, two different shapes depending on the container.

Cooking oil takes the shape of the bottle in which it is stored.



Same cooking oil, different shapes in different bottles.

Air expands to fill an inflated balloon.



Same air, expanded to fill the entire space inside the balloon.

3. Fluids Exert Pressure



Every fluid pushes against surfaces that come into contact with it.



This pushing force is called **fluid pressure**.



A fluid exerts pressure not only downward because of gravity but also sideways and, in some cases, upward.

Examples include:



Water pushing against the walls of a swimming pool.



Air pressing against the inside of a basketball.



Water spraying out of holes in the side of a bottle.



Divers feeling greater pressure as they descend deeper into the ocean.

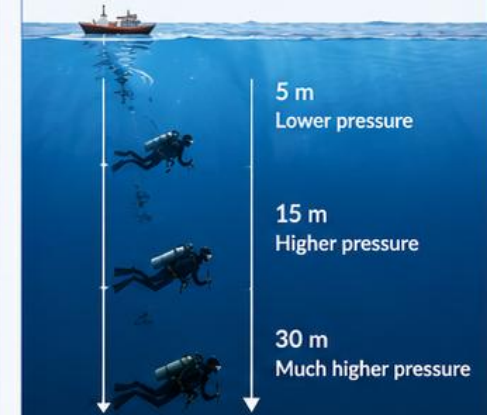


Fluid pressure acts in all directions.



Pressure increases with depth.

The deeper you go in a fluid, the greater the pressure.



Fluid pressure is one of the most important concepts in fluid physics because it explains how hydraulic systems function and why water behaves differently at different depths.



Hydraulic systems use fluid pressure to lift, push, and move heavy loads.



Water pressure increases with depth, which is why dams are built thicker at the bottom.

4. Fluids Can Transmit Pressure



One remarkable property of fluids—particularly liquids—is their ability to **transmit applied pressure** throughout the fluid.

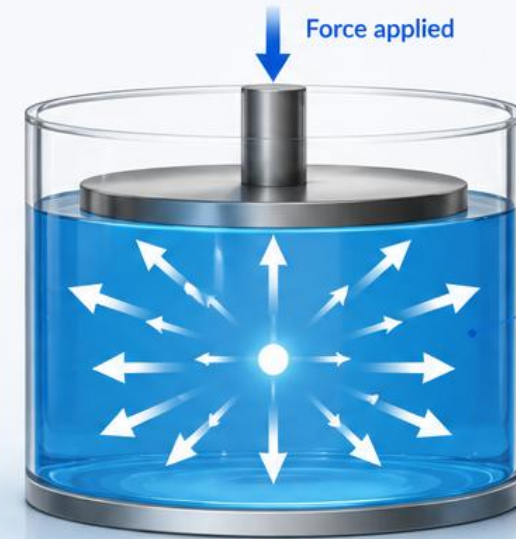


When pressure is applied to a confined liquid, that pressure **spreads throughout the liquid**.



This behavior forms the basis of **Pascal's Principle**, which will be studied later.

When pressure is applied to a confined liquid, it is transmitted equally in all directions throughout the fluid and to the walls of the container.



Pressure is transmitted equally throughout the liquid.

Applications include:

Hydraulic car lifts



Lift heavy vehicles with a small applied force.

Hydraulic brakes



Convert a small pedal force into strong braking power.

Excavators



Use hydraulic systems to move large loads and dig with powerful force.

Barber chairs



Hydraulic lifts allow smooth height adjustment.

Dentist chairs



Hydraulic systems position the chair for patient comfort and treatment.



These machines allow a relatively **small applied force** to help move or lift **much heavier loads**.

5. Fluids Exert an Upward Buoyant Force



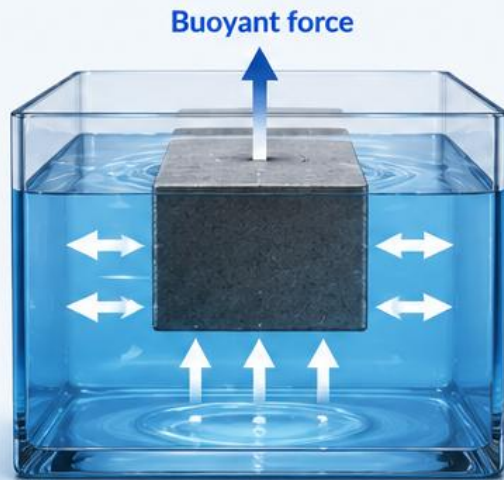
When an object is placed in a fluid, the fluid **pushes upward** on the object.



This upward force is called **buoyancy** or the **buoyant force**.



Buoyancy determines whether an object **floats**, **sinks**, or remains **suspended** in the fluid.



Fluid pressure is greater at greater depths.

The difference in pressure creates an upward force.



This behavior is explained by **Archimedes' Principle**, which will be discussed in later lessons.

Examples:



Boats float on rivers.

Buoyant force supports the boat and keeps it afloat.



Ships travel across oceans.

Their large volume of fluid displaced provides the lift they need to float.



Ice cubes float in water.

Ice is less dense than liquid water, so it floats.



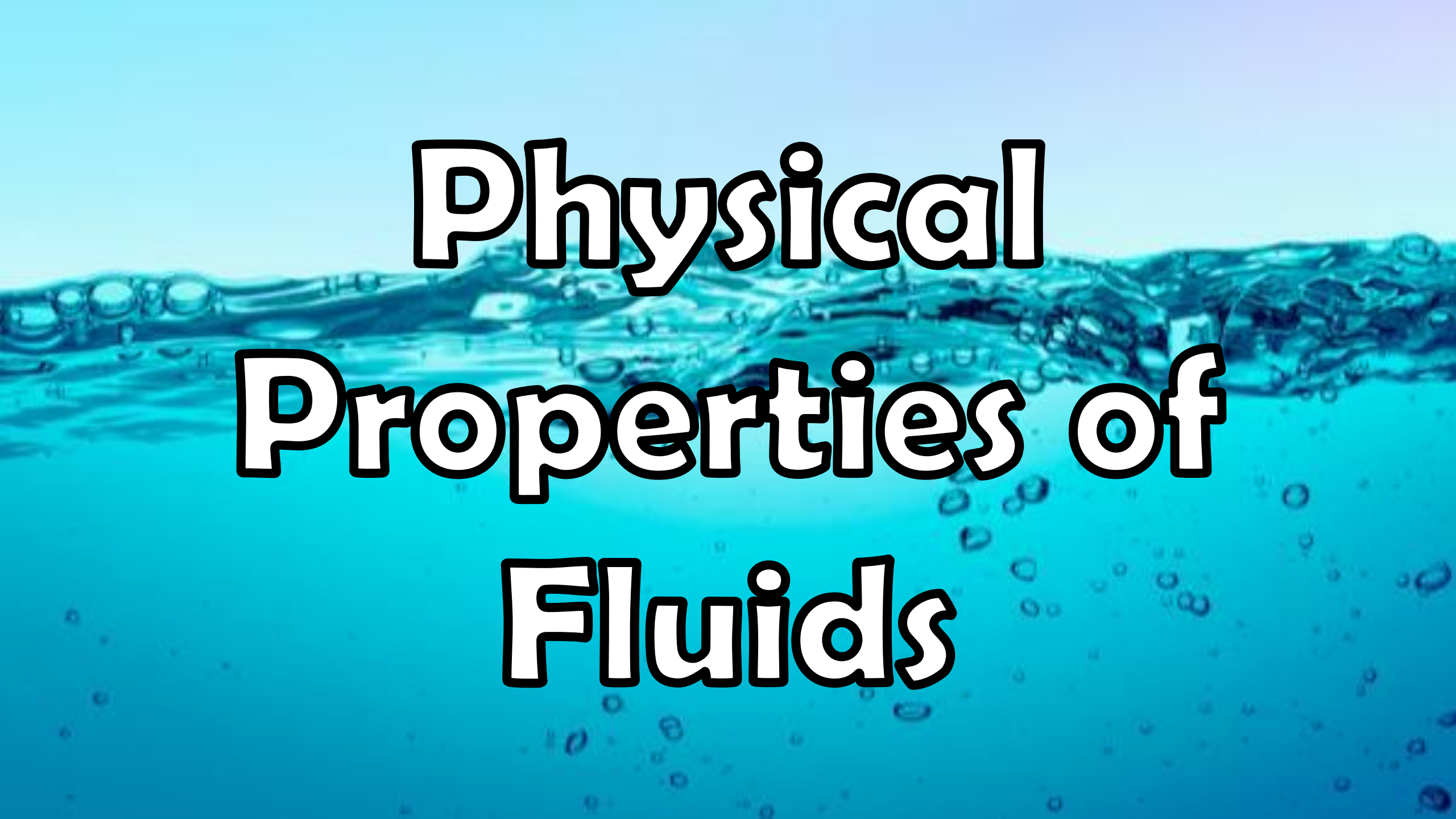
Life jackets help people stay afloat.

They provide buoyant force that keeps people above water.



Submarines can sink or rise by adjusting their buoyancy.

They take in or release water to change their average density.



Physical Properties of Fluids

Density



Density describes how much matter is **packed** into a given amount of space.



Some fluids have **higher density** than others.

Examples:



Honey is **denser** than water.

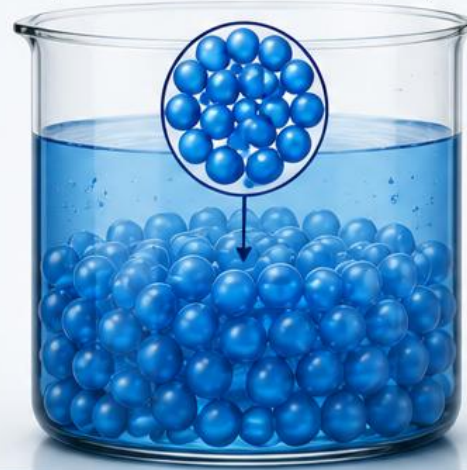


Cooking oil is **less dense** than water, which is why oil floats on top of water.

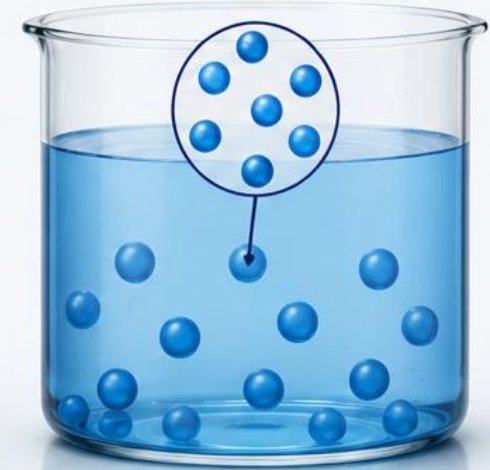
More dense

Less dense

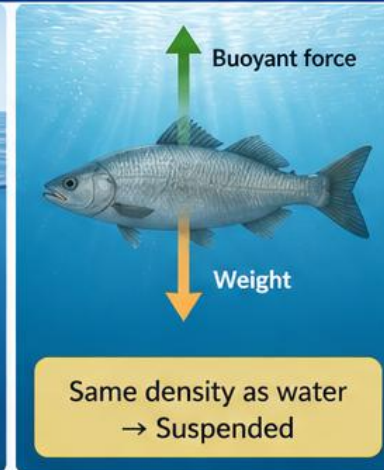
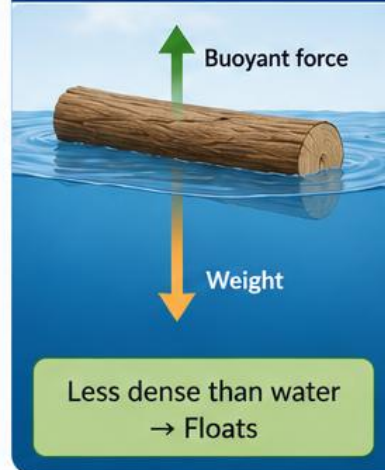
High density
(more matter in the same space)



Low density
(less matter in the same space)



Density plays a major role in determining whether objects **float** or **sink**.



It is also one of the most important concepts in understanding **Archimedes' Principle**.

Viscosity

Viscosity is a fluid's **resistance to flowing**.



A highly viscous fluid flows slowly because its particles resist movement.



This property explains why honey pours much more slowly than water.

Examples:

High viscosity
(flows slowly)



Honey



Syrup



Molasses

Low viscosity
(flows easily)



Water



Alcohol



Gasoline

HIGH VISCOSITY (flows slowly)



LOW VISCOSITY (flows easily)



VS.

VISCOSITY COMPARISON

More viscous
(flows slower)

Less viscous
(flows faster)



Viscosity is important in selecting lubricants for engines.



Viscosity is important in designing pipelines for the efficient transport of fluids.

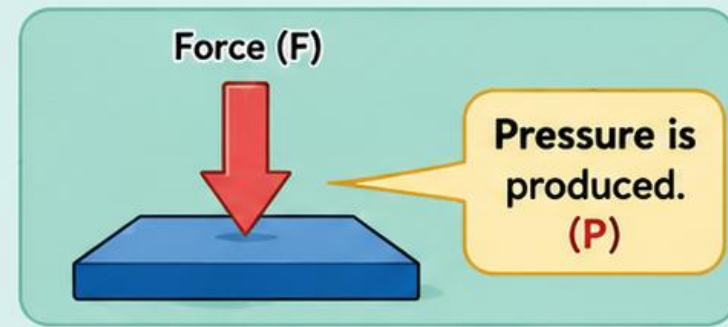


Viscosity is important in producing food products with the right texture and consistency.



WHAT IS PRESSURE?

Pressure is the effect of a **force** acting on a **surface**.



--- EXAMPLES IN OUR DAILY LIFE ---

1 Pressing your finger on a sponge



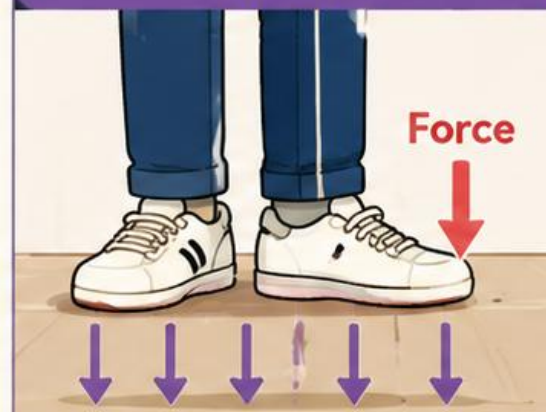
Your finger pushes down.
The sponge feels the push.
Pressure is produced.

2 Pushing a marker against paper



The marker pushes on the paper.
The paper feels the push.
Pressure is produced.

3 Standing on the floor



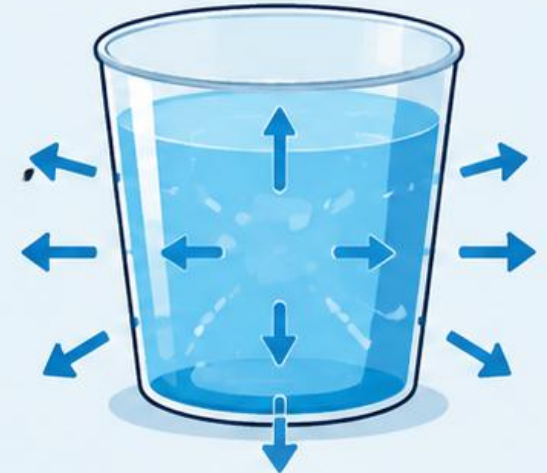
Your weight pushes down
on the floor.
Pressure is produced.



Whenever a force pushes on something,
pressure is produced.

FOR FLUIDS...

Pressure exists
inside the
liquid or gas.



In a fluid, pressure acts in
all directions.

EVERYDAY EXAMPLES OF FLUID PRESSURE



When force is applied to a confined fluid, the **pressure spreads** throughout and pushes in **all directions** until it reaches the **only opening**.

1 Toothpaste Tube



When you squeeze the middle,



pressure is created inside the toothpaste,



the pressure **spreads** through the paste,



toothpaste moves toward the **only opening**.



The toothpaste does not know “where to go.” It simply moves toward the **easiest escape**.

2 Juice Pouch



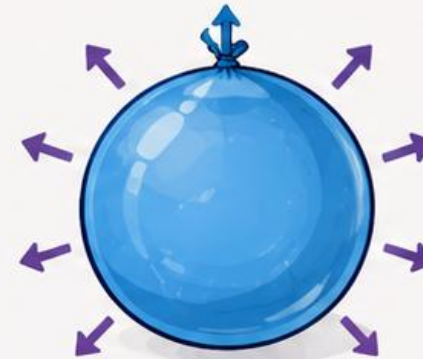
When squeezed, the juice experiences **pressure throughout** the pouch...

...and comes out **through the straw**.

3 Water Balloon



Press one side of a water balloon.



Notice that another side bulges outward.

The **pressure** is transmitted **throughout** the water.

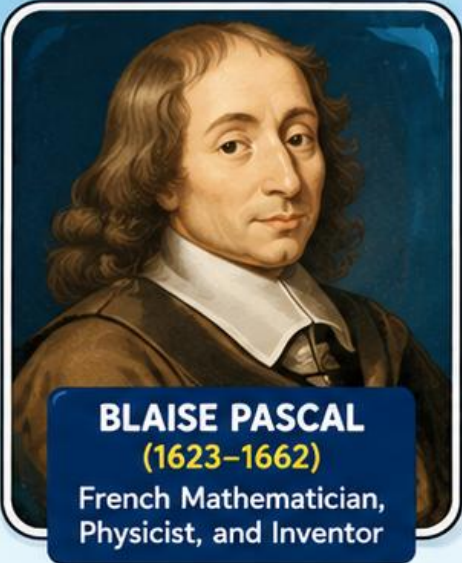


Fluids (like paste, juice, or water) **cannot be easily compressed**, so the push you apply is **transmitted throughout** the fluid and to **every part** of its container.



PASCAL'S PRINCIPLE

PRESSURE APPLIED
to one part of
the confined fluid.



BLAISE PASCAL
(1623–1662)

French Mathematician,
Physicist, and Inventor



French scientist
Blaise Pascal
discovered an
important idea
about fluids.

PASCAL'S PRINCIPLE

“ When **pressure** is applied
to a **confined fluid**,
the **pressure** is transmitted
equally and undiminished
in all directions throughout
the fluid. ”



The **pressure**
does not
weaken or
decrease.

It is transmitted
undiminished.



Liquids are
almost impossible
to compress,
so they **transmit**
pressure very
effectively.



Push **one part** of a trapped liquid, and **every part** of the liquid experiences that pressure.

VISUALIZING PASCAL'S PRINCIPLE



Pascal's Principle: When pressure is applied to a confined fluid, the pressure is transmitted **equally** and **undiminished** in all directions throughout the fluid.

1

Imagine a **sealed** syringe filled completely with **water**.



2

If you **push** the plunger,



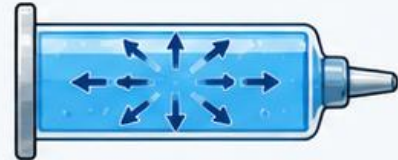
3

the **water pushes equally** against

- the walls,
- the bottom,
- and every part inside the syringe.

4

The **pressure** spreads everywhere.

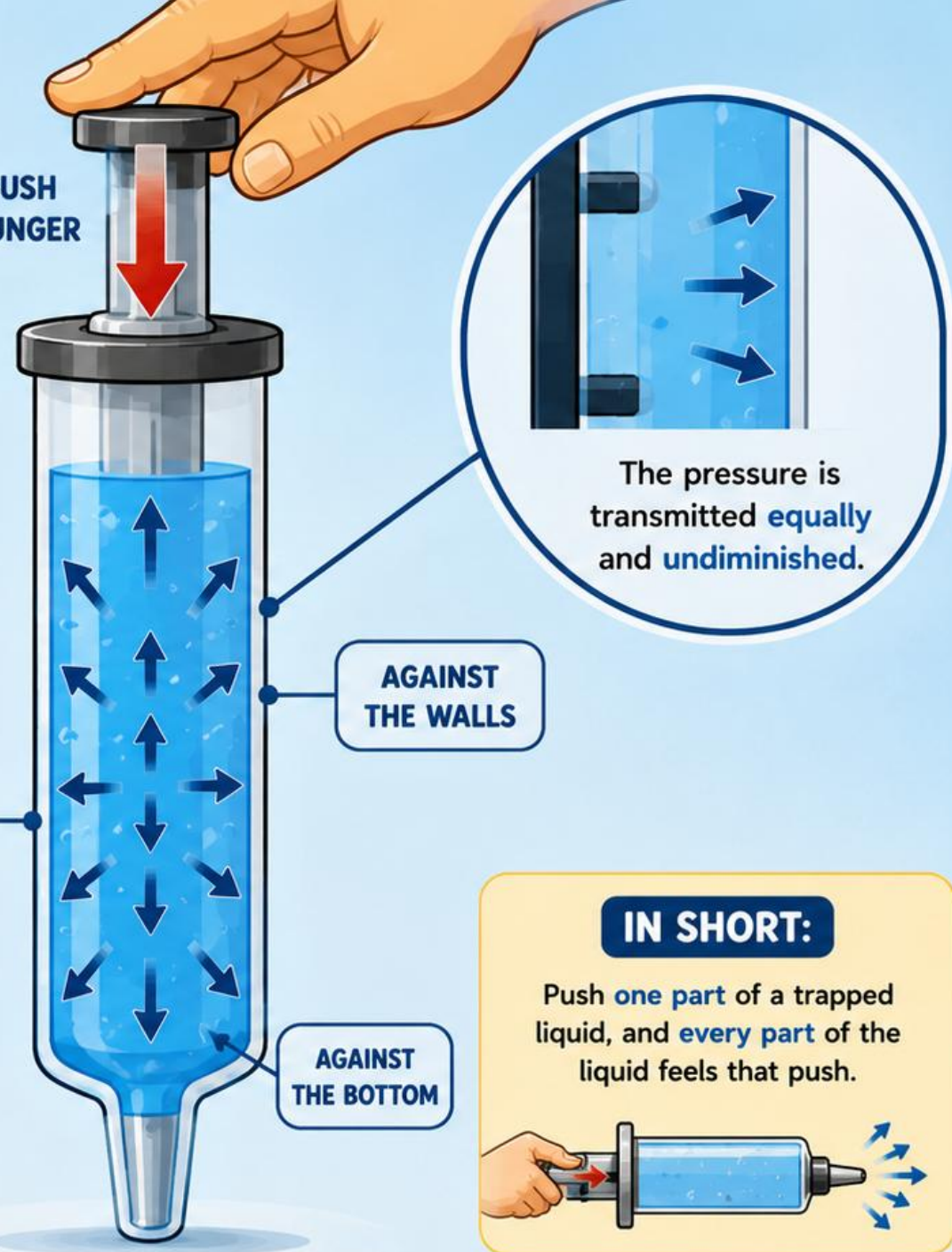


5

The liquid acts like a **messenger** carrying your push throughout the container.



YOU PUSH
THE PLUNGER



IN SHORT:

Push **one part** of a trapped liquid, and **every part** of the liquid feels that push.



REAL-LIFE APPLICATIONS OF PASCAL'S PRINCIPLE

1 HYDRAULIC CAR LIFT

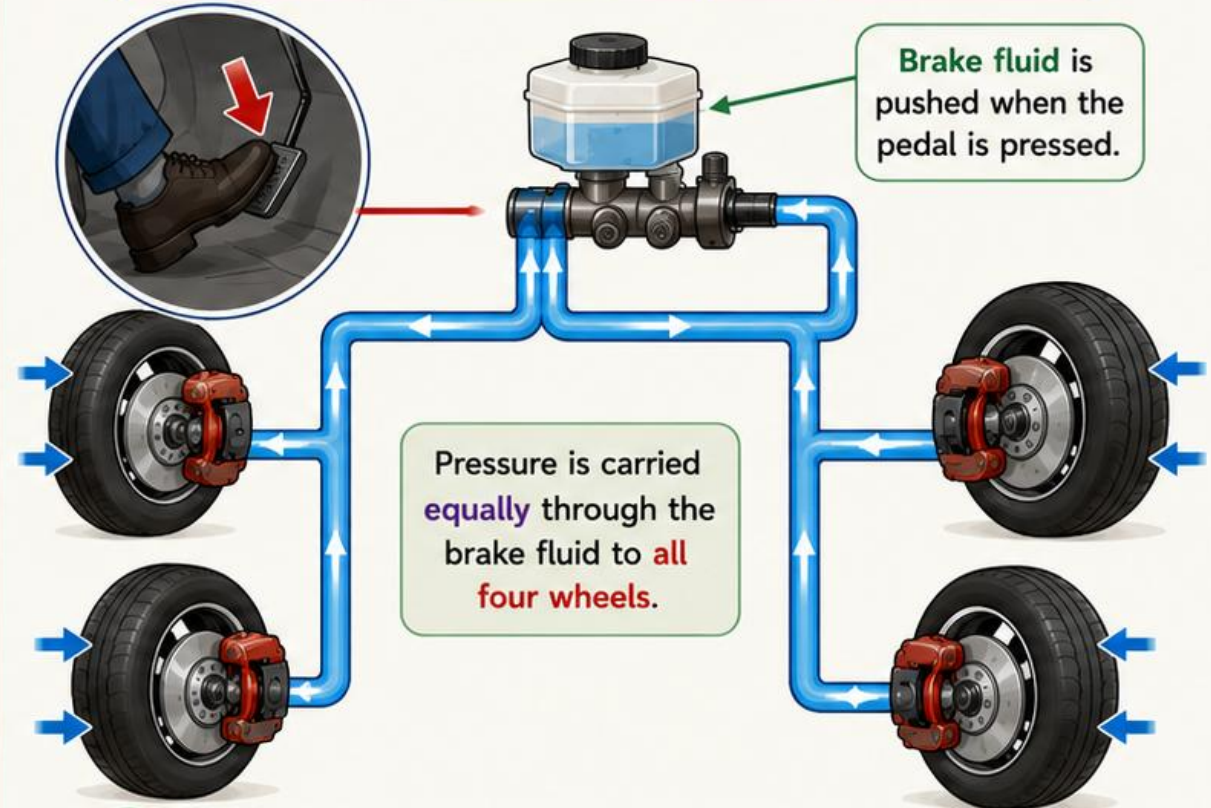
- A mechanic pushes a **small piston**.
- The **liquid** transfers the **pressure**.
- A **larger piston** lifts the entire car.



A **small force** on a small piston can lift a **heavy load** on a **larger piston**!

2 HYDRAULIC BRAKES

- When the driver presses the **brake pedal**,
- the **brake fluid** carries the **pressure**
- to **all four wheels** simultaneously.
- This allows the vehicle to **stop safely**.



Equal pressure = **Balanced braking** = **Safe stopping**

HAVE YOU EVER WONDERED
WHY A **HUGE SHIP**
MADE OF STEEL
CAN **FLOAT** WHILE
A SMALL **STONE SINKS?**

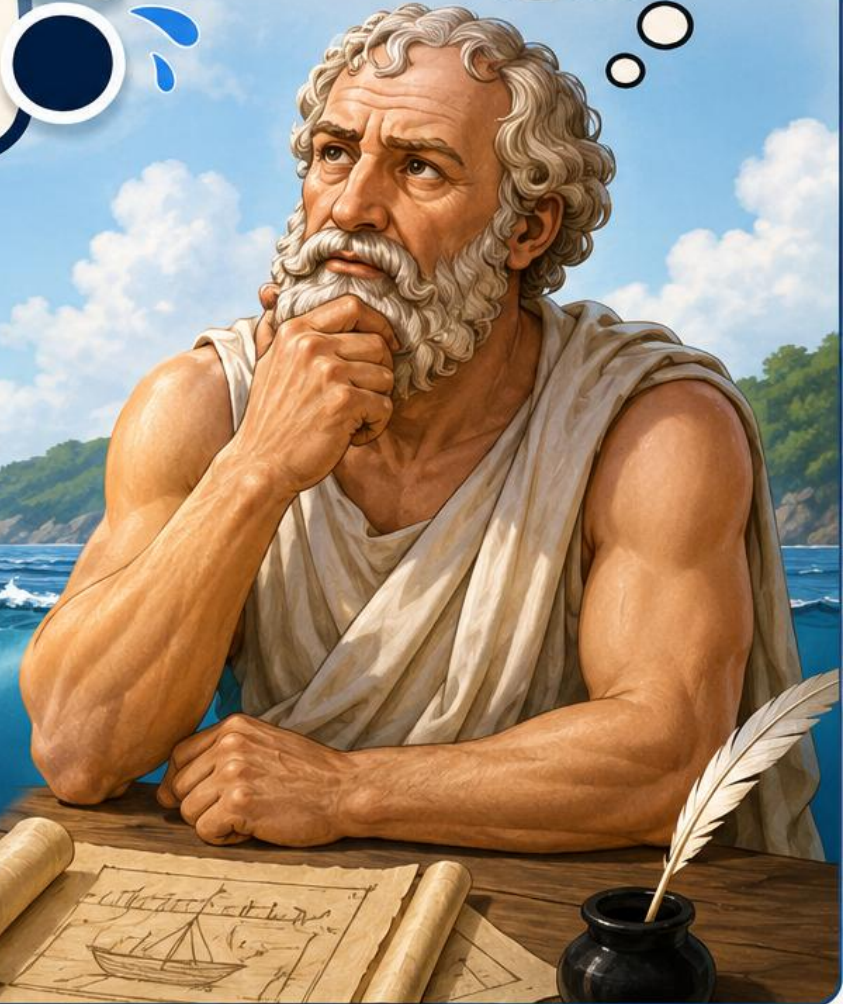


ARCHIMEDES

FLOATS

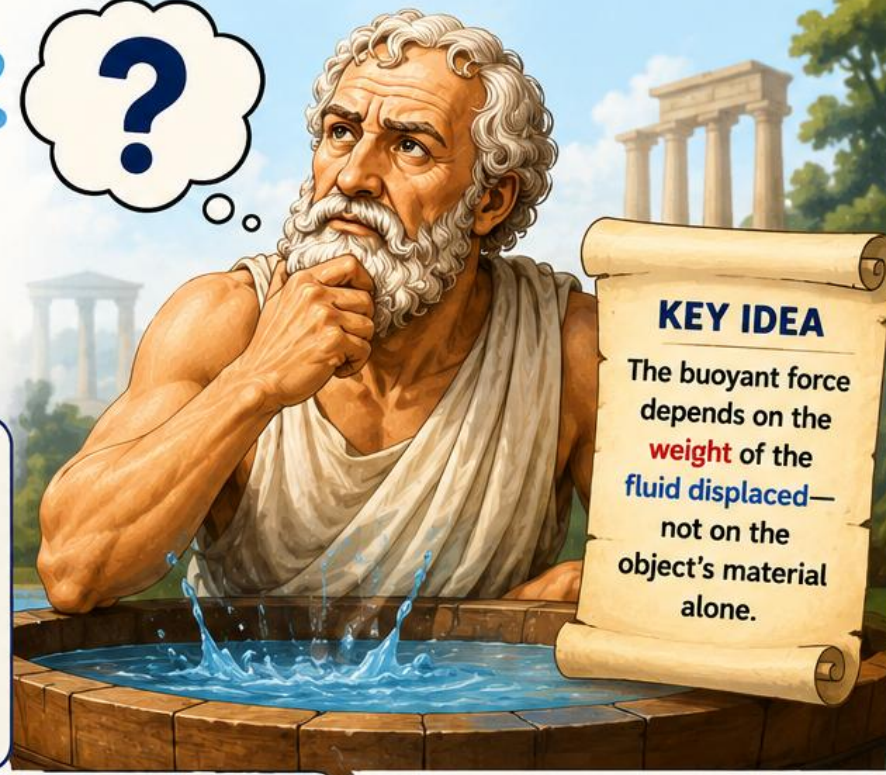


SINKS





WHAT IS ARCHIMEDES' PRINCIPLE?



KEY IDEA

The buoyant force depends on the **weight** of the **fluid displaced**—not on the object's material alone.

ARCHIMEDES' PRINCIPLE STATES:

When an object is partially or completely immersed in a **fluid**, it experiences an **upward force** called the **buoyant force**. This buoyant force is **equal** to the weight of the **fluid displaced** by the object.



HOW IT WORKS

When an object is placed in a fluid, it pushes some of the fluid aside. The fluid pushes back **upward** with an **equal force**.



Object enters the fluid.

It pushes the fluid aside.

The fluid pushes back upward with an equal force.

EXAMPLES

✓ FLOATS

A large steel ship floats because it displaces a large amount of water, creating enough **buoyant force** to support its weight.



✗ SINKS

A small stone sinks because it displaces only a small amount of water, producing a **buoyant force** that is less than its weight.



THE FORMULA

Buoyant Force (F_b)
=
Weight of the Displaced Fluid (W_f)

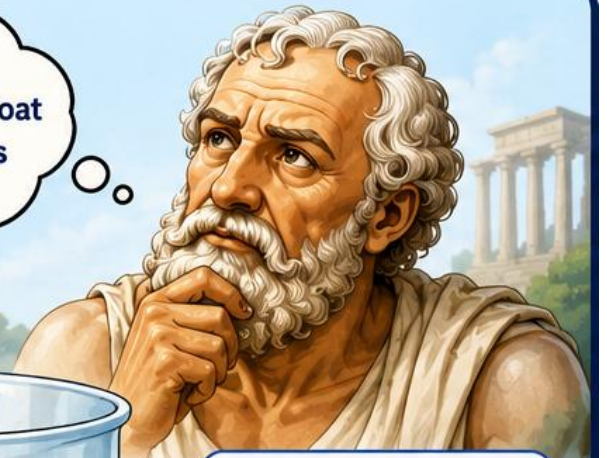
$$F_b = W_f$$



This is why a huge steel ship can **float** while a small stone **sinks**.

UNDERSTANDING BUOYANT FORCE

Why do some things float while others sink?



Buoyant force is the **upward force** exerted by a fluid (such as water or air) on an object placed in it.



This upward force **acts against** the object's weight.



BUOYANT FORCE

The fluid pushes the object **UPWARD**.

WEIGHT

Gravity pulls the object **DOWNWARD**.

WHAT HAPPENS?

FLOATS

If the buoyant force is **greater than or equal to** the object's weight, the object **floats**.



SINKS

If the buoyant force is **less than** the object's weight, the object **sinks**.



KEY IDEA



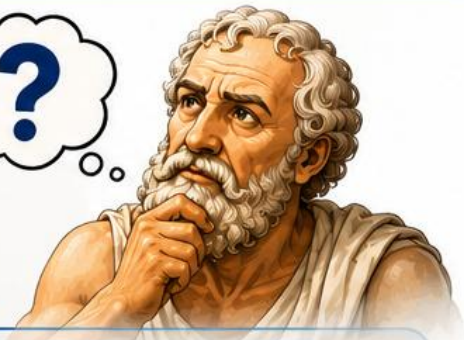
The buoyant force depends on the **weight of the fluid displaced**, not on the object's material alone.



The **more fluid** an object displaces, the **greater** the buoyant force.



THREE POSSIBLE OUTCOMES:



An object in a fluid can experience one of three outcomes depending on the relationship between **buoyant force** and **weight**.

1

FLOAT



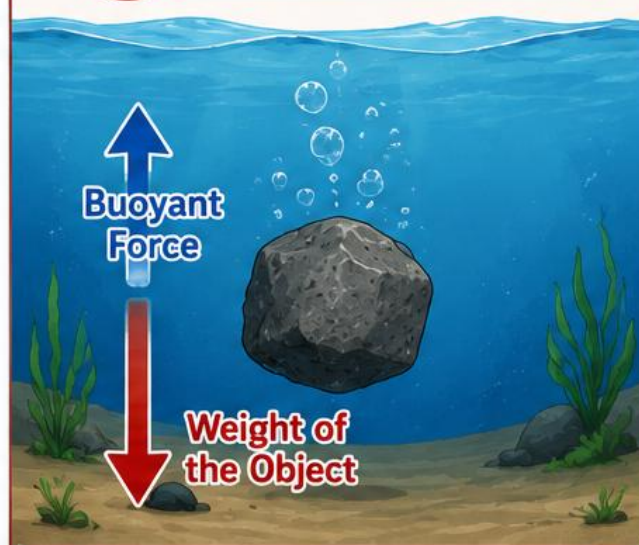
Buoyant force is **greater than or equal to** the object's weight.



The object **FLOATS**.

2

SINK



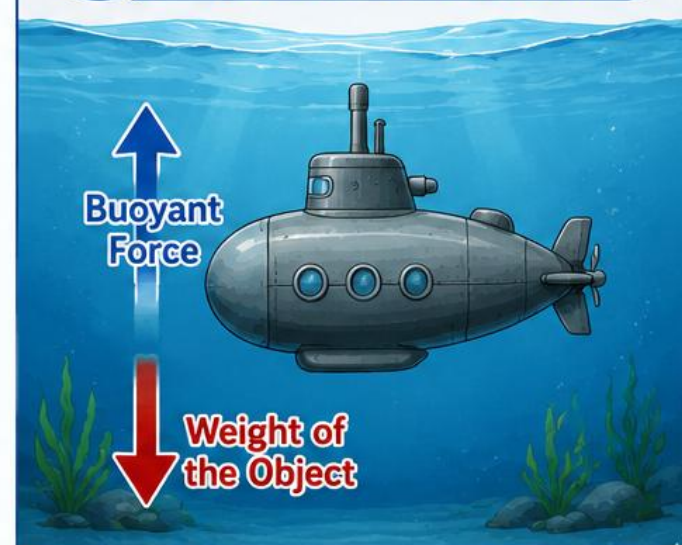
Object's weight is **greater than** the buoyant force.



The object **SINKS**.

3

STAY SUSPENDED
(NEUTRAL BUOYANCY)



Buoyant force **equals** the object's weight.



The object stays **SUSPENDED**.



The balance between **buoyant force** and **weight** determines the outcome!

[https://youtu.be/0eIXMs_NQVU?
si=GbFjVmZkh1WTQfAo](https://youtu.be/0eIXMs_NQVU?si=GbFjVmZkh1WTQfAo)